

CE 356 — Hydraulic Engineering

Pipe and Pipe Networks: Multiple-Choice Practice Questions

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Instructions: Each question has four options (A–D). Select the single best answer. An answer key with brief explanations is provided at the end of this booklet. Questions are grouped by topic in the order the material was presented in class.

Section 1: Background, Continuity & Energy Equations

1. In pipe systems under pressure flow (e.g. water distribution systems, sewer interceptors), problems can generally be solved using which two fundamental equations?
 - A) Bernoulli's equation and Newton's second law
 - B) The continuity equation and the steady-state energy equation
 - C) Manning's equation and the momentum equation
 - D) The Colebrook equation and Darcy's law
2. The continuity equation between two points in a pipe system is written as $Q_1 = A_1V_1 = Q_2 = A_2V_2$. What physical principle does this express?
 - A) Conservation of energy along the pipe
 - B) Conservation of mass (volumetric flow rate is constant)
 - C) Conservation of momentum at a junction
 - D) Conservation of pressure head
3. In the steady-state energy equation $Z_1 + P_1/\gamma + V_1^2/2g = Z_2 + P_2/\gamma + V_2^2/2g + h_f + h_m$, the term P/γ represents:
 - A) Velocity head
 - B) Pressure head
 - C) Elevation head
 - D) Total head
4. In the steady-state energy equation, h_f and h_m respectively represent:
 - A) Elevation head and pressure head
 - B) Friction (major) losses and minor losses
 - C) Pump head and turbine head
 - D) Hydraulic grade line and energy grade line
5. A total head of 'zero' pressure and velocity head but positive elevation head at a free water surface of a reservoir is consistent with which boundary condition commonly used in pipe-flow problems?
 - A) The reservoir surface is treated as open to atmosphere with negligible velocity, so $P/\gamma = 0$ and $V^2/2g \approx 0$
 - B) The reservoir must always be pressurized
 - C) The reservoir elevation is irrelevant to the energy equation
 - D) Velocity head at a reservoir surface is always maximum

Section 2: Hydraulic Grade Line (HGL) and Energy Grade Line (EGL)

6. The Hydraulic Grade Line (HGL) at any point along a pipe represents:
 - A) Elevation head only
 - B) Pressure head plus elevation head
 - C) Velocity head plus elevation head
 - D) Total head (elevation + pressure + velocity head)
7. The Energy Grade Line (EGL) at any point along a pipe represents:
 - A) Pressure head plus elevation head
 - B) Elevation head only
 - C) The total head — elevation + pressure + velocity head
 - D) The friction slope only
8. For a uniform pipe (constant diameter, so $V_1 = V_2$), the relationship between EGL and HGL is:
 - A) EGL is always above HGL by a varying amount
 - B) EGL is parallel to HGL, separated by a constant velocity head
 - C) EGL and HGL coincide exactly
 - D) HGL is always above EGL
9. The slope of the energy grade line, S_f , is defined as:
 - A) $S_f = h_f \times L$
 - B) $S_f = h_f / L$
 - C) $S_f = L / h_f$
 - D) $S_f = h_f / V$
10. According to the lecture, what causes discontinuities (abrupt drops) in the EGL along a pipeline?
 - A) Changes in pipe material color
 - B) Minor losses at fittings, and the presence of pumps or turbines
 - C) Only elevation changes in the pipe
 - D) Changes in fluid temperature
11. For a pump located along a pipeline, how is its effect incorporated into the energy equation?
 - A) An energy term is subtracted from the left-hand side (LHS)
 - B) An energy term is added to the left-hand side (LHS)
 - C) It has no effect on the energy equation
 - D) It is added to both sides equally, cancelling out
12. For a turbine located along a pipeline, how is its effect incorporated into the energy equation?
 - A) An energy term is added to the LHS
 - B) An energy term is subtracted from the LHS
 - C) It is ignored since turbines don't affect head
 - D) It is added to the RHS as a loss term identical to h_f

Section 3: Major (Friction) Losses — Chezy, Darcy-Weisbach, Reynolds Number

13. Which THREE equations are named in the slides as means of computing major (friction) head loss, h_f ?
 - A) Bernoulli, Euler, and Navier-Stokes equations
 - B) Chezy equation, Darcy-Weisbach equation, and Hazen-Williams equation
 - C) Continuity, momentum, and energy equations
 - D) Colebrook, Jain, and Moody equations only

14. The Chezy equation is given as $V = C\sqrt{RS}$. Substituting $S = h_f/L$ and solving for h_f gives:
- A) $h_f = LV^2/(RC^2) = 4LV^2/(dC^2)$
 - B) $h_f = RC^2/(LV^2)$
 - C) $h_f = L/(RC^2V^2)$
 - D) $h_f = LV/(RC)$
15. The Darcy-Weisbach equation for friction head loss is:
- A) $h_f = f(L/d)(V^2/2g)$
 - B) $h_f = C\sqrt{RS}$
 - C) $h_f = 0.849 CH R^{0.63} S f^{0.54}$
 - D) $h_f = 6.35 n^2 LV^2/D^{4/3}$
16. In the Darcy-Weisbach equation $h_f = f(L/d)(V^2/2g)$, the variables f , L , d , V , and g stand respectively for:
- A) Flow rate, length, density, viscosity, gravity
 - B) Friction factor, pipe length, pipe diameter, velocity, and gravitational acceleration
 - C) Force, load, drag, volume, and gauge pressure
 - D) Frequency, lift, drop, viscosity, and gradient
17. Equating the Darcy-Weisbach and Chezy formulas gives the relationship between the friction factor f and the Chezy coefficient C as:
- A) $f = 8g/C^2$ (or $C = \sqrt{8g/f}$)
 - B) $f = C^2/8g$
 - C) $f = 8gC$
 - D) $f = C/8g$
18. The Reynolds number for pipe flow is defined as $Re = \rho Vd/\mu$. What does Re primarily indicate?
- A) The pipe roughness relative to diameter
 - B) Whether the flow is laminar or turbulent
 - C) The exact value of the friction factor without further calculation
 - D) The total head loss in the pipe
19. In the Darcy-Weisbach framework, for laminar flow ($Re < 2000$), the friction factor f is given simply by:
- A) $f = 0.079/Re^{0.25}$
 - B) $f = 64/Re$
 - C) $f = 0.25/[\log(k/3.7D + 5.74/Re^{0.9})]^2$
 - D) f depends only on pipe roughness
20. For turbulent flow ($Re > 4000$), the Darcy-Weisbach friction factor f depends on:
- A) Reynolds number only
 - B) Pipe roughness only
 - C) Both Reynolds number and relative pipe roughness (k/D)
 - D) Neither Reynolds number nor roughness — it is constant
21. Pipe roughness for the Darcy-Weisbach approach is characterized using:
- A) The Manning roughness coefficient, n , only
 - B) The Hazen-Williams coefficient, CH , only
 - C) The equivalent sand roughness, ϵ (or k)
 - D) The Reynolds number

Section 4: Moody Diagram, Colebrook and Jain Equations

22. The Moody diagram, developed from experiments in the 1930s–1940s, graphically expresses the relationship between:

- A) Hazen-Williams coefficient and pipe diameter
- B) Friction factor (f) and Reynolds number (Re), for various relative roughness values
- C) Head loss and pipe length only
- D) Pump power and flow rate

23. The Colebrook equation, $1/\sqrt{f} = -2\log(k/D/3.7 + 2.51/(Re\sqrt{f}))$, is notable because:

- A) It gives f explicitly with no iteration required
- B) It is an implicit equation for f and must be solved iteratively
- C) It only applies to laminar flow
- D) It ignores relative roughness entirely

24. According to the lecture, friction factors predicted by the Colebrook equation are generally accurate to within:

- A) 1-2% of experimental data
- B) 10-15% of experimental data
- C) 50% of experimental data
- D) Exactly 0% error

25. The Jain equation, $1/\sqrt{f} = -2\log(ks/3.7D + 5.74/Re^{0.9})$, was derived by Jain in 1976 primarily to:

- A) Replace the concept of Reynolds number entirely
- B) Provide an explicit empirical relationship approximating the Moody diagram curves, avoiding the need to iterate as with Colebrook
- C) Compute minor losses instead of major losses
- D) Determine the Hazen-Williams coefficient

26. The Jain equation is valid within which stated ranges?

- A) $10^{-5} \leq ks/D \leq 10^{-2}$ and $5,000 \leq Re \leq 10^8$
- B) $0 \leq ks/D \leq 1$ and $Re < 2000$ only
- C) Any ks/D and any Re
- D) $ks/D > 1$ and $Re > 10^8$

27. As a rule of thumb given in the lecture, what accuracy in calculating friction losses in pipes should generally be expected (given uncertainties in relative roughness and underlying data)?

- A) About 1%
- B) About 10%
- C) About 50%
- D) About 90%

Section 5: Hazen-Williams Equation and Manning's Formula

28. The Hazen-Williams equation $V = 0.849 CH R^{0.63} Sf^{0.54}$ is commonly used for:

- A) Compressible gas flow in pipes
- B) Pipe flow in water-supply systems
- C) Open channel flow only
- D) Laminar flow in capillary tubes

29. In the Hazen-Williams equation, R is the hydraulic radius, computed for a circular pipe as:

- A) $R = d$
- B) $R = d/2$
- C) $R = d/4$
- D) $R = 4/d$

30. The Hazen-Williams equation for head loss is $h_f = 6.82 (L/D^{1.17})(V/CH)^{1.85}$. The Hazen-Williams equation is stated to be applicable to water at approximately what temperature, pipe diameter range, and velocity range?

- A) 100°C; any diameter; any velocity
- B) 16°C; diameters between 50 mm and 1850 mm; velocities less than 3 m/s
- C) 0°C; diameters over 5 m; velocities greater than 10 m/s
- D) 16°C; only diameters below 10 mm; any velocity

31. For circular pipes, substituting $V = Q/A$, $A = \pi d^2/4$, and $R = d/4$ into the Hazen-Williams velocity equation yields the metric discharge form:

- A) $Q = 0.278 CH d^{2.63} S^{0.54}$
- B) $Q = 0.849 CH d^{0.63} S^{0.54}$
- C) $Q = 6.82 CH d^{1.17} S$
- D) $Q = CH \sqrt{(dS)}$

32. In the Hazen-Williams roughness table, a 'new, unlined' ductile/cast iron pipe has a typical CH value of about:

- A) 80
- B) 110
- C) 130
- D) 150

33. An 'old, unlined' ductile/cast iron pipe typically has a much lower Hazen-Williams coefficient than a new pipe because:

- A) Old pipes are always larger in diameter
- B) Internal corrosion/tuberculation increases roughness over time, reducing CH
- C) CH values increase with pipe age due to smoothing
- D) Age has no effect on CH

34. Manning's formula for velocity in a closed conduit is $V = (1/n) R^{(2/3)} S^{(1/2)}$. This formula is best suited for:

- A) Laminar, low-Reynolds-number flow only
- B) Fully turbulent flows, where frictional head losses depend primarily on relative roughness
- C) Flows where roughness is irrelevant
- D) Compressible flow of gases

35. The Manning-based head loss formula given in the lecture is $h_f = 6.35 n^2 LV^2/D^{(4/3)}$. Here ' n ' represents:

- A) The number of pipes in the network
- B) The Manning roughness coefficient
- C) The Hazen-Williams coefficient
- D) The kinematic viscosity

36. In the worked example (150 mm new ductile iron pipe, $V = 1$ m/s, $L = 500$ m), the Hazen-Williams method gave $h_f \approx 3.85$ m, while the Manning method gave $h_f \approx 6.73$ m. This illustrates that:

- A) The two empirical head-loss formulas can give noticeably different results for the same conditions
- B) Manning's and Hazen-Williams formulas always give identical results
- C) Head loss is independent of the formula chosen
- D) Only Darcy-Weisbach can be used for ductile iron pipe

Section 6: Applying Darcy-Weisbach — Classes of Problems & Minor Losses

37. The Darcy-Weisbach equation $h_f = f(L/d)(V^2/2g)$ has three main variables: h_f , d , and V (or Q). The three main classes of problems identified in the lecture are:

- A) Compute h_f given d and V ; compute V given d and h_f ; compute d given h_f and Q
- B) Compute f given h_f and d only
- C) Compute L given h_f and V only
- D) Compute g given d and V only

38. To compute h_f directly when d and V are both known, the correct procedure is:

- A) Guess h_f , then iterate
- B) Determine Re from V , d , and μ ; determine f from Moody diagram/Jain equation; then compute h_f directly from the D-W equation
- C) Assume $f = 1$ always
- D) Use the Hazen-Williams equation only

39. When solving for velocity V given pipe diameter d and head loss h_f (V and f both unknown), the lecture recommends:

- A) A trial-and-error procedure: assume f , compute V from the D-W equation, determine Re and hence f , and iterate until f converges
- B) Solving directly with no iteration since V can be isolated algebraically together with f
- C) Ignoring friction entirely
- D) Using only the laminar flow formula regardless of Re

40. When solving for pipe diameter d given head loss h_f and flow rate Q , the recommended trial-and-error approach involves:

- A) Assuming f , computing d from the D-W equation combined with continuity ($V = Q/A$), determining Re and hence f , and iterating until f converges
- B) Assuming d directly with no reference to f or Re
- C) Ignoring continuity
- D) Solving explicitly for d with a single calculation

41. Minor losses in a piping system arise due to fittings such as valves, bends, elbows, tees, inlets, exits, enlargements, and contractions. Physically, these losses are caused by:

- A) Only pipe wall friction along straight sections
- B) Flow separation and mixing at the fitting
- C) Temperature changes in the fluid
- D) Reduction in fluid density

42. Minor losses are computed using the relation $h_m = K(V^2/2g)$. The loss coefficient K is:

- A) The same fixed value ($K = 1.0$) for all fittings
- B) A function of the type of fitting, and assumed independent of Reynolds number
- C) Only applicable to laminar flow
- D) Always negative

43. From the minor loss coefficient table, which fitting has by far the largest loss coefficient (K_m), effectively acting almost like a full blockage?

- A) A gate valve fully open ($K_m = 0.15$)
- B) A swing check valve in backward flow ($K_m = \text{infinity}$)
- C) A long-radius 90° flanged elbow ($K_m = 0.2$)
- D) A rounded pipe entrance ($K_m = 0.2$)

44. From the minor loss coefficients for pipe entrances, which entrance type has the LOWEST loss coefficient?

- A) Re-entrant (pipe juts into tank), $K = 1.0$
- B) Square connection, $K = 0.5$
- C) Rounded connection, $K = 0.2$
- D) Sharp-edged, $K = 0.5$

Section 7: Pipes in Series and Parallel

45. For pipes connected in series (a compound pipeline), the governing relationships are:

- A) $Q_1 = Q_2 = Q_3$ (same discharge throughout) and $h_f = h_{f1} + h_{f2} + h_{f3} + \dots$ (head losses add up)
- B) $Q_1 + Q_2 + Q_3 = \text{constant}$ and $h_{f1} = h_{f2} = h_{f3}$
- C) $Q_1 = Q_2 = Q_3$ and $h_f = h_{f1} = h_{f2} = h_{f3}$
- D) Pressure is identical at every point regardless of pipe diameter changes

46. For pipes connected in parallel between two common points, the governing relationships are:

- A) $Q = Q_1 + Q_2 + Q_3$ and $h_f = h_{f1} = h_{f2} = h_{f3}$
- B) $Q = Q_1 = Q_2 = Q_3$ and $h_f = h_{f1} + h_{f2} + h_{f3}$
- C) $Q_1 = Q_2 = Q_3$ only, with no relation for head loss
- D) $h_{f1} \neq h_{f2} \neq h_{f3}$ always, with no restriction

47. In the worked series-pipe example (15 cm pipe, 6 m long, followed by 22.5 cm pipe, 16 m long, with a 6 m head difference between reservoirs, $f = 0.04$), which losses were included in the head loss summation?

- A) Only friction loss in the first pipe
- B) Entrance loss, friction loss in pipe 1, sudden enlargement loss, friction loss in pipe 2, and exit loss (velocity head)
- C) Only the exit loss
- D) Only losses due to pumps

48. In the parallel-pipe example (pipe network between points A and B, with two parallel pipes of different length/diameter, head loss = 2 m, $f = 0.01$), the total discharge Q_1 (main pipe) was found by:

- A) Subtracting the two branch flows
- B) Adding the two branch flows ($Q_{\text{main}} = Q_{\text{branch1}} + Q_{\text{branch2}}$), since head loss is equal in each branch but flows differ
- C) Taking the average of the two branch flows
- D) Ignoring one of the branches entirely

Section 8: Pipelines with Pumps and Turbines

49. The energy equation for a pipeline including a pump (adding head h_A) and a turbine (removing head h_R) is written as:

- A) $Z_1 + P_1/\gamma + V_1^2/2g = Z_2 + P_2/\gamma + V_2^2/2g + h_L$ (no pump/turbine terms)
- B) $Z_1 + P_1/\gamma + V_1^2/2g + h_A - h_R = Z_2 + P_2/\gamma + V_2^2/2g + h_L$
- C) $Z_1 + P_1/\gamma + V_1^2/2g - h_A + h_R = Z_2 + P_2/\gamma + V_2^2/2g + h_L$
- D) $Z_1 + P_1/\gamma + V_1^2/2g = Z_2 + P_2/\gamma + V_2^2/2g - h_L$

50. The power delivered by a pump is given by $P = \gamma Q h_A$. If $\gamma = 9.81 \text{ kN/m}^3$, Q is in m^3/s , and h_A (head added) is in m, then P is expressed in units of:

- A) Newtons (N)
- B) kilowatts (kW)
- C) Pascals (Pa)
- D) m^3/s

51. Rearranging $P = \gamma Q h_A$, the head added by a pump of known power P_A and flow Q is given by:

- A) $h_A = P_A/(Q\gamma)$
- B) $h_A = P_A Q \gamma$
- C) $h_A = \gamma/(P_A Q)$
- D) $h_A = Q\gamma/P_A$

Section 9: Pipe Networks — Configurations and Water Transmission

52. Water transmission refers to the transportation of water from the source to the treatment plant and to the area of distribution. It can be realized through:

- A) Free-flow (gravity) conduits only
- B) Pressurized pipelines (pumping) only
- C) Free-flow conduits, pressurized pipelines, or a combination of the two
- D) Only open-channel flow, never pipes

53. A water supply distribution system, as described in the lecture, consists of a complex network of:

- A) Interconnected pipes, service reservoirs, and pumps
- B) Only pipes, with no reservoirs or pumps needed
- C) Only treatment plants
- D) Only pumps and valves, with no pipe network

54. Because water demand is highly variable (by day and season) while supply is normally constant, the distribution system must:

- A) Ignore variability and rely purely on pump speed changes
- B) Include storage elements and be capable of flexible operation
- C) Operate only at night when demand is lowest
- D) Use only branching (dead-end) networks

55. Water pressures within a distribution system are normally kept above a minimum head of about:

- A) 1 m
- B) 5 m
- C) 15 m
- D) 50 m

- 56.** The two main pipe network configurations discussed are:
- A) Series and parallel systems
 - B) Branching system and grid (ring/loop) system
 - C) Laminar and turbulent systems
 - D) Open and closed conduit systems
- 57.** The structure of a 'branching' type water distribution network is described as being similar to:
- A) A closed loop
 - B) A tree, with a main supply line and branch lines
 - C) A perfect circle
 - D) A single straight pipe
- 58.** The ring (grid/loop) system is generally preferred over the branching system in water distribution mainly because:
- A) It is always cheaper to construct
 - B) It prevents 'dead ends' (reducing stagnant water risk) and permits more flexible operation, especially during repairs
 - C) It requires fewer pipes overall
 - D) It eliminates the need for any pumps
- 59.** According to the lecture, the components of a complete water supply system include:
- A) Only the treatment plant
 - B) Source, treatment plant, storage tanks/reservoirs, and water transmission/distribution
 - C) Only pumps and valves
 - D) Only the distribution mains, with no treatment step

Section 10: Pipe Network Analysis and the Hardy-Cross Method

- 60.** Pipe network analysis involves determining pipe flow rates and pressure heads at outflow points such that they satisfy:
- A) Only the continuity equation
 - B) Only the energy equation
 - C) Both the continuity and energy equations
 - D) Neither equation — only empirical judgment
- 61.** Which THREE analysis methods for pipe networks are named in the lecture?
- A) Hardy-Cross Method, Nodal Method, and Newton-Raphson Method
 - B) Chezy Method, Manning Method, and Colebrook Method
 - C) Bernoulli Method, Euler Method, and Laplace Method
 - D) Moody Method, Jain Method, and Reynolds Method
- 62.** The Hardy-Cross method's continuity condition, applied at any junction/node, requires:
- A) Total inflow to equal total outflow at the node
 - B) Total inflow to always exceed total outflow
 - C) Flow to reverse direction at every node
 - D) All connecting pipes to have equal diameters
- 63.** The Hardy-Cross energy condition states that around any closed loop in the network:
- A) The algebraic sum of head losses is zero: $\sum h_l(\text{loop}) = 0$, i.e., $\sum K(Q + \Delta Q)^n = 0$
 - B) The sum of head losses must equal the pump head exactly
 - C) Head losses must all be positive with no sign convention
 - D) The sum of flow rates around the loop must equal zero, ignoring head loss

64. A pipe network, per the lecture, must satisfy three basic conditions. Which of the following is NOT one of them?

- A) At any junction, total inflow must equal total outflow
- B) The algebraic sum of head losses around any closed circuit is zero
- C) The head loss equation must be satisfied for each pipe
- D) All pipes in the network must have identical lengths

65. Both the Darcy-Weisbach and Hazen-Williams head loss formulas can be written in the general resistance form $h_f = KQ^n$. For Darcy-Weisbach expressed as $h_f = (16/\pi^2) \cdot f \cdot (L/d^5) \cdot (Q^2/2g)$, the exponent n equals:

- A) 1
- B) 1.85
- C) 2
- D) 2.63

66. In the Hazen-Williams-based resistance form $h_f = (4.727L / (C^{1.85} d^{4.87})) Q^{1.85}$, the exponent n used in Hardy-Cross calculations is:

- A) 1.85
- B) 2.0
- C) 2.63
- D) 4.87

67. In the Hardy-Cross method, the flow correction for a loop is computed as $\delta = -\sum h_f(\text{loop}) / [n \sum |h_f/Q|(\text{loop})]$. Here 'n' represents:

- A) The number of pipes in the loop
- B) The exponent in the resistance formula (e.g., 2 for Darcy-Weisbach, 1.85 for Hazen-Williams)
- C) The Manning roughness coefficient
- D) The number of loops in the entire network

68. In the Hardy-Cross correction formula, why does the denominator contain $|h_f/Q|$ (absolute value) rather than simply h_f/Q ?

- A) To avoid sign cancellation issues; the denominator should always add up as a positive magnitude regardless of assumed flow direction
- B) Because head loss is never negative in any pipe
- C) It is purely a typographical convention with no mathematical reason
- D) To make the correction term always negative

69. In the Hardy-Cross procedure, when a pipe is a common member between two adjacent loops, how is its correction handled?

- A) The pipe is ignored in one of the loops
- B) Both loop corrections are applied to it: as a member of loop 1, $\delta = \delta_1 - \delta_2$; as a member of loop 2, $\delta = \delta_2 - \delta_1$
- C) Only the larger correction is applied
- D) The pipe's flow is fixed and never corrected

70. The overall Hardy-Cross procedure begins by dividing the network into closed loops, then for each loop: assuming discharge and direction (clockwise positive), calculating equivalent pipe resistance K , computing $hf = KQa^n$ retaining sign, and summing Σhf for the loop. What is the next key step?

- A) Stop — the assumed flows are already the final answer
- B) Calculate $|hf/Qa|$ for each pipe, sum for the loop, then compute correction $\delta = -\Sigma hf / (n \Sigma |hf/Qa|)$ and apply it to update Qa
- C) Immediately solve for nodal pressures without further iteration
- D) Discard the loop and start the network calculation over from scratch

71. The Hardy-Cross iterative process for a loop is considered converged (correction step complete) when:

- A) δ becomes very small and $\Sigma hf \approx 0$ for each loop
- B) Only one iteration has been performed, regardless of the result
- C) Q for every pipe becomes exactly zero
- D) The number of loops equals the number of pipes

72. After the flows in a pipe network have converged via Hardy-Cross, the final step of the procedure is to:

- A) Discard the results and restart with a different assumed flow
- B) Solve for the pressure (or pressure head) at each node using energy conservation
- C) Recompute Q values without regard to any prior convergence
- D) Ignore pressures entirely, since only flow rates matter

Section 11: Worked Numerical Examples

73. In the worked example: water at velocity 1.7 m/s flows through a 150 mm pipe ($Q = 0.03 \text{ m}^3/\text{s}$, $L = 500 \text{ m}$, roughness = 0.06 mm, water at 10°C , $\mu = 1.30 \times 10^{-3} \text{ N}\cdot\text{s}/\text{m}^2$). The Reynolds number was calculated as approximately:

- A) 1.96×10^2
- B) 1.96×10^3
- C) 1.96×10^4
- D) 1.96×10^5

74. In that same example, using the Moody diagram with $Re \approx 1.96 \times 10^4$ and relative roughness $k/D = (0.06 \times 10^{-3})/0.15 = 4 \times 10^{-4}$, the friction factor f was read as approximately:

- A) 0.001
- B) 0.018
- C) 0.18
- D) 1.8

75. Continuing that example, using the Darcy-Weisbach equation with $f = 0.018$, $L = 500 \text{ m}$, $V = 1.7 \text{ m/s}$, $D = 0.15 \text{ m}$, the computed head loss hf was approximately:

- A) 0.884 m
- B) 8.84 m
- C) 88.4 m
- D) 884 m

76. Given $h_f \approx 8.84$ m for the pipe in that example, the corresponding pressure drop $\Delta P = \rho g h_f$ was calculated as approximately:

- A) 867 Pa
- B) 8,670 Pa
- C) 86,700 Pa (8.67×10^4 Pa)
- D) 867,000 Pa

77. In the two-reservoir example (height difference 15 m, pipeline 350 mm diameter, 1000 m long, $f = 0.005$, ignoring all local losses), applying $15 = f L Q^2 / (3d^5)$ gave the discharge Q as approximately:

- A) 21.7 L/s
- B) 217 L/s
- C) 2170 L/s
- D) 0.0217 L/s

78. In the 61 m cast-iron pipeline example (two reservoirs, 6 m elevation difference, $Q = 0.06$ m³/s at 15.6°C, entrance/exit/valve/bend losses included), applying the energy equation between the two reservoir surfaces (both at atmospheric pressure, negligible velocity) directly gave the total head loss, h_{loss} , as:

- A) 0 m
- B) 6 m
- C) 15 m
- D) 61 m

79. In the same 61 m cast-iron pipeline example, the sum of minor loss coefficients (entrance 0.5 + exit 1.0 + globe valve 10.0 + two 90° bends 1.8) totalled:

- A) 3.3
- B) 13.3
- C) 23.3
- D) 33.3

80. In the pipe-diameter-comparison example (20 cm diameter, 45 m — note: length label shown as 'l' in formula uses 45 m — pipe, $V = 2.5$ m/s), using $h_f = f l V^2 / (2dg)$ with $f = 0.0242$ and using $V = C \sqrt{RS}$ with $C = 57$, both methods yielded a head loss of approximately:

- A) 0.173 m
- B) 1.73 m
- C) 17.3 m
- D) 173 m

81. In the parallel-pipe example (pipe 1: $L = 25$ m, $d = 0.04$ m; pipe 2: $L = 30$ m, $d = 0.05$ m; head loss between A and B = 2 m; $f = 0.01$), the velocity in pipe 1 was found to be approximately:

- A) 0.25 m/s
- B) 2.5 m/s
- C) 25 m/s
- D) 0.025 m/s

82. In that same parallel-pipe example, the total discharge in the main pipe (Q_1 branch + Q_2 branch) was found to be approximately:

- A) 0.0817×10^3 m³/s
- B) 0.817×10^3 m³/s
- C) 8.17×10^3 m³/s
- D) 81.7×10^3 m³/s

83. In the square-loop Hardy-Cross example (all pipes 1 km long, 300 mm diameter, $f = 0.0163$, minor losses neglected), the equivalent resistance constant K' in $HL = K'Q^2$ was computed to be approximately:

- A) 5.54
- B) 55.4
- C) 554
- D) 5540

84. In the first trial of that square-loop example (assumed flows: $AB = 60$, $BC = 40$, $CD = 0$, $AD = -40$ L/s, clockwise positive), the sum of head losses around the loop, ΣHL , was calculated as 2.00 m, which is greater than the 0.01 m convergence tolerance. The corresponding flow correction ΔQ was calculated as approximately:

- A) +12.92 L/s
- B) -12.92 L/s
- C) -1.292 L/s
- D) -129.2 L/s

85. After applying the correction $\Delta Q \approx -12.92$ L/s to all four pipes in the square-loop example and recomputing, the second-trial ΣHL was approximately -0.0107 m, which was judged acceptable. The final (rounded) discharge in pipe AB was:

- A) 13 L/s
- B) 27 L/s
- C) 47 L/s
- D) 53 L/s

Answer Key & Explanations

Section 1: Background, Continuity & Energy Equations

1. Correct answer: B

Explanation: The lecture states that pipe flow problems are solved using the continuity equation and the steady-state energy equation.

2. Correct answer: B

Explanation: Continuity expresses conservation of mass — for an incompressible fluid, volumetric flow rate Q is the same at every cross-section.

3. Correct answer: B

Explanation: P/γ is the pressure head; Z is elevation head, $V^2/2g$ is velocity head, and $h_f + h_m$ are the head losses.

4. Correct answer: B

Explanation: h_f denotes friction (major) losses along the pipe, and h_m denotes minor losses due to fittings, valves, bends, etc.

5. Correct answer: A

Explanation: At a large reservoir surface, gauge pressure is zero (atmospheric) and the surface velocity is essentially zero, so only the elevation head term remains.

Section 2: Hydraulic Grade Line (HGL) and Energy Grade Line (EGL)

6. Correct answer: B

Explanation: HGL = pressure head + elevation head ($P/\gamma + Z$).

7. Correct answer: C

Explanation: EGL depicts the elevation of the total head ($Z + P/\gamma + V^2/2g$) along the pipe.

8. Correct answer: B

Explanation: When $V_1 = V_2$ (uniform pipe), the velocity head is constant, so EGL runs parallel to HGL, offset by $V^2/2g$.

9. Correct answer: B

Explanation: $S_f = h_f/L$, i.e., head loss due to friction per unit length of pipe.

10. Correct answer: B

Explanation: The EGL has discontinuities at fittings due to minor losses, and pumps/turbines also introduce discontinuities.

11. Correct answer: B

Explanation: For a pump, energy is being added to the flow, so an energy term (h_A) is added to the LHS of the energy equation.

12. Correct answer: B

Explanation: A turbine removes energy from the flow, so its energy term (h_R) is subtracted from the LHS.

Section 3: Major (Friction) Losses — Chezy, Darcy-Weisbach, Reynolds Number

13. Correct answer: B

Explanation: The slide 'Major Losses' explicitly lists the Chezy equation, Darcy-Weisbach equation, and Hazen-Williams equation.

14. Correct answer: A

Explanation: As shown in the slide: $hf = LV^2/RC^2 = 4LV^2/dC^2$ (using $R = d/4$ for a circular pipe).

15. Correct answer: A

Explanation: $hf = f \cdot (L/d) \cdot (V^2/2g)$ is the Darcy-Weisbach equation, where f is the (dimensionless) friction factor.

16. Correct answer: B

Explanation: f = friction factor [-], L = pipe length [L], d = pipe diameter [L], V = velocity [LT^{-1}], g = gravity [LT^{-2}].

17. Correct answer: A

Explanation: As derived on the slide: $f = 8g/C^2$, equivalently $C = \sqrt{8g/f}$.

18. Correct answer: B

Explanation: Reynolds number characterizes flow conditions (laminar vs turbulent) based on the ratio of inertial to viscous forces.

19. Correct answer: B

Explanation: For laminar flow, $f = 64/Re$ — a simple function of Reynolds number alone.

20. Correct answer: C

Explanation: For turbulent flow, f is a function of both Re and the pipe's relative roughness, typically read from the Moody diagram or computed via Colebrook/Jain equations.

21. Correct answer: C

Explanation: Pipe roughness is determined from the equivalent sand roughness, denoted ϵ or k (e.g., ks in mm).

Section 4: Moody Diagram, Colebrook and Jain Equations

22. Correct answer: B

Explanation: The Moody diagram plots friction factor f against Reynolds number Re for a family of curves representing different relative roughness (ks/D) values.

23. Correct answer: B

Explanation: The Colebrook equation has f appearing on both sides (implicitly), so it must be solved iteratively (or via the Moody diagram).

24. Correct answer: B

Explanation: The slide states Colebrook-predicted friction factors are generally accurate to within 10–15% of experimental data.

25. Correct answer: B

Explanation: Jain's equation offers an explicit (non-iterative) approximation to the implicit Colebrook relationship for the Moody diagram family of curves.

26. Correct answer: A

Explanation: As stated on the slide, the Jain equation applies for $10^{-5} \leq ks/D \leq 10^{-2}$ and $5,000 \leq Re \leq 10^8$.

27. Correct answer: B

Explanation: The lecture notes an accuracy of about 10% in calculating friction losses in pipes is to be expected due to uncertainties in relative roughness and the Colebrook data.

Section 5: Hazen-Williams Equation and Manning's Formula

28. Correct answer: B

Explanation: The Hazen-Williams relationship is commonly used specifically for pipe flow in water-supply systems.

29. Correct answer: C

Explanation: For a circular pipe flowing full, hydraulic radius $R = d/4$.

30. Correct answer: B

Explanation: The slide states the Hazen-Williams equation applies to water at 16°C, in pipes 50 mm–1850 mm diameter, with flow velocities less than 3 m/s.

31. Correct answer: A

Explanation: The metric Hazen-Williams discharge formula is $Q = 0.278 CH d^{2.63} S^{0.54}$.

32. Correct answer: C

Explanation: From Table 2.2, new unlined ductile/cast iron pipe has CH range 120–140, typical value 130.

33. Correct answer: B

Explanation: As pipes age, internal roughness (e.g., corrosion, tuberculation) increases, which is reflected by a decrease in the Hazen-Williams coefficient (typical value ~80 for old unlined iron vs 130 for new).

34. Correct answer: B

Explanation: The lecture states the Manning formula applies to fully turbulent flows where frictional head losses depend primarily on relative roughness.

35. Correct answer: B

Explanation: n is the Manning roughness coefficient (a property of the pipe material).

36. Correct answer: A

Explanation: The example explicitly compares (a) Hazen-Williams and (b) Manning results for the same pipe/flow, showing the empirical formulas can yield different head-loss estimates.

Section 6: Applying Darcy-Weisbach — Classes of Problems & Minor Losses

37. Correct answer: A

Explanation: The three classes are: (1) compute h_f given d and V ; (2) compute V given d and h_f ; (3) compute d given h_f and Q .

38. Correct answer: B

Explanation: Since V and d are known, Re can be computed directly, f found from Moody/Jain, and h_f then computed directly — no iteration required.

39. Correct answer: A

Explanation: Since Re depends on the unknown V , an iterative trial-and-error approach (assume f , solve for V , recompute Re and f) is required.

40. Correct answer: A

Explanation: As d and V are both unknown (and V depends on the unknown d via continuity), the solution again requires iterative trial and error.

41. Correct answer: B

Explanation: Minor losses are caused by flow separation and mixing induced by fittings interrupting the smooth flow of fluid.

42. Correct answer: B

Explanation: K is a function of fitting type (different for each fitting) and is generally assumed independent of Re .

43. Correct answer: B

Explanation: A swing check valve in backward flow has $K_m = \text{infinity}$, since it is designed to block reverse flow entirely.

44. Correct answer: C

Explanation: A rounded (well-rounded) entrance connection has the lowest loss coefficient ($K = 0.2$ in the pipe-entrance table; well-rounded general entrance can be as low as 0.04).

Section 7: Pipes in Series and Parallel

45. Correct answer: A

Explanation: In series, the same discharge flows through each pipe (continuity), while the total head loss equals the sum of the individual head losses.

46. Correct answer: A

Explanation: In parallel, the total flow splits among the pipes ($Q = Q_1 + Q_2 + Q_3$), but the head loss is identical along each parallel path since they share the same start and end points.

47. Correct answer: B

Explanation: The solution sums: entrance loss (h_{m1}), friction in pipe 1 (h_{f1}), sudden enlargement loss (h_{m2}), friction in pipe 2 (h_{f2}), and the exit/velocity head loss — all expressed in terms of $v^2/2g$.

48. Correct answer: B

Explanation: Since both parallel branches experience the same head loss (2 m) but can carry different discharges, the total flow in the main pipe is the sum of the two branch discharges.

Section 8: Pipelines with Pumps and Turbines

49. Correct answer: B

Explanation: Pump head h_A is added and turbine head h_R is subtracted on the LHS: $Z_1 + P_1/\gamma + V_1^2/2g + h_A - h_R = Z_2 + P_2/\gamma + V_2^2/2g + h_L$.

50. Correct answer: B

Explanation: $P = \gamma Q h_A$ with γ in kN/m^3 and Q in m^3/s and h in m gives P in $\text{kN}\cdot\text{m}/\text{s} = \text{kW}$.

51. Correct answer: A

Explanation: As stated on the slide: $h_A = P_A/(Q\gamma)$ (and similarly $h_R = P_R/(Q\gamma)$ for a turbine).

Section 9: Pipe Networks — Configurations and Water Transmission

52. Correct answer: C

Explanation: The lecture states water transmission can use free-flow (gravity) conduits, pressurized pipelines (pumping), or a combination of both.

53. Correct answer: A

Explanation: The system consists of interconnected pipes, service reservoirs, and/or pumps, delivering water from the treatment plant to the consumer.

54. Correct answer: B

Explanation: To reconcile variable demand with constant supply, the system must include storage elements (reservoirs/tanks) and allow flexible operation.

55. Correct answer: C

Explanation: The lecture states water pressures are normally kept above a minimum of about 15 m head.

56. Correct answer: B

Explanation: The two main configurations are the branching system (tree-like) and the grid/ring/loop system.

57. Correct answer: B

Explanation: The branching system has a tree-like structure, consisting of a main supply line with branch lines extending from it.

58. Correct answer: B

Explanation: Loop/grid systems avoid dead ends (reducing stagnant-water risk) and allow more flexible operation, particularly when sections need repair, since water can reach a point via multiple paths.

59. Correct answer: B

Explanation: The listed components are: (1) Source, (2) Treatment plant, (3) Storage tanks/reservoirs, (4) Water transmission/distribution.

Section 10: Pipe Network Analysis and the Hardy-Cross Method

60. Correct answer: C

Explanation: Pipe network analysis requires flow rates and pressure heads that simultaneously satisfy both continuity and energy equations.

61. Correct answer: A

Explanation: The three analysis methods listed are: (1) Hardy-Cross Method (looped method), (2) Nodal Method, (3) Newton-Raphson Method.

62. Correct answer: A

Explanation: At every node/junction, inflow must equal outflow (e.g., $Q_a = Q_b + Q_c$, as shown for node A in the diagram).

63. Correct answer: A

Explanation: The energy equation condition is that the algebraic sum of head losses around any closed loop equals zero.

64. Correct answer: D

Explanation: Identical pipe lengths is not a stated requirement. The three real conditions are: inflow=outflow at junctions, zero net head loss around loops, and the head-loss equation holding for each pipe.

65. Correct answer: C

Explanation: The Darcy-Weisbach form shown has Q^2 , so $n = 2$ (compare to Hazen-Williams, $h_f = (4.727L/(C^{1.85} d^{4.87}))Q^{1.85}$, where $n = 1.85$).

66. Correct answer: A

Explanation: For the Hazen-Williams form, $n = 1.85$ (the exponent on Q).

67. Correct answer: B

Explanation: In this formula, n is the exponent from the general resistance relationship $h_f = KQ^n$ (not the number of pipes or loops).

68. Correct answer: A

Explanation: Since h_f carries the sign of the assumed flow direction, taking the absolute value in the denominator ensures a consistently positive sum, so the numerator's sign properly determines the direction of correction.

69. Correct answer: B

Explanation: For a pipe shared between two loops, both loop corrections apply with appropriate signs: $\delta = \delta_1 - \delta_2$ (as seen from loop 1) and $\delta = \delta_2 - \delta_1$ (as seen from loop 2).

70. Correct answer: B

Explanation: After computing Σh_f , the next step is computing $\Sigma |h_f/Q_a|$, deriving the correction δ , applying $Q_{\text{new}} = Q_a + \delta$, and repeating until Σh_f becomes negligibly small (i.e., converges).

71. Correct answer: A

Explanation: Iterations stop when the correction δ becomes very small and the algebraic sum of head losses around each loop is approximately zero.

72. Correct answer: B

Explanation: The final listed step is to solve for pressure at each node using conservation of energy, once the flow distribution has converged.

Section 11: Worked Numerical Examples

73. Correct answer: C

Explanation: $Re = \rho V d / \mu = (1000 \times 1.7 \times 0.15) / (1.30 \times 10^{-3}) \approx 1.96 \times 10^5$, confirming turbulent flow ($Re > 2000$).

74. Correct answer: B

Explanation: The example states $f \approx 0.018$ was read from the Moody diagram for these Re and k/D values.

75. Correct answer: B

Explanation: $h_f = f(L/D)(V^2/2g) = 0.018 \times (500/0.15) \times (1.7^2)/(2 \times 9.81) \approx 8.84$ m, as shown in the worked solution.

76. Correct answer: C

Explanation: $\Delta P = \rho g h_f = 1000 \times 9.81 \times 8.84 \approx 8.67 \times 10^4$ Pa, as given in the worked solution.

77. Correct answer: B

Explanation: Solving $15 = 317.33Q^2$ gives $Q = \sqrt{0.0473} \approx 0.217$ m³/s = 217 L/s, as shown in the worked solution.

78. Correct answer: B

Explanation: Since $Z_1 - Z_2 = 20 - 0$ (with all other terms zero at the reservoir surfaces, per the given elevations), the total loss works out to $h_{\text{loss}} = 6$ m — matching the given 6 m elevation difference between the reservoirs.

79. Correct answer: B

Explanation: $0.5 + 1.0 + 10.0 + 1.8 = 13.3$, matching the 'Total' row in the minor-losses table of the worked example.

80. Correct answer: B

Explanation: Both the Darcy-type formula ($f = 0.0242$) and the Chezy-based formula ($C = 57$) yielded $h_f \approx 1.73$ m, demonstrating the two equivalent approaches give consistent results.

81. Correct answer: B

Explanation: Solving $0.01 \times (25/0.04) \times V_1^2 / (2 \times 9.81) = 2$ gives $V_1 \approx 2.506$ m/s, close to 2.5 m/s.

82. Correct answer: C

Explanation: $Q_{\text{branch1}} \approx 3.15 \times 10^{-3}$ m³/s and $Q_{\text{branch2}} \approx 5.02 \times 10^{-3}$ m³/s, summing to about 8.17×10^{-3} m³/s for the combined (main) pipe flow.

83. Correct answer: C

Explanation: Substituting the given L , D , and f into $HL = f(L/D)(V^2/2g)$ and expressing V in terms of Q/A gives $HL = 554Q^2$, so $K' \approx 554$.

84. Correct answer: B

Explanation: $\Delta Q = -\Sigma HL / (2\Sigma|HL/Q|) = -2 / (2 \times 0.0774) \approx -12.92$ L/s.

85. Correct answer: C

Explanation: The final rounded discharges given were: $AB = 47$ L/s, $BC = 27$ L/s, $CD = -13$ L/s, $AD = -53$ L/s.